

FINFLOW ANALYTICS

Real-Time SaaS Fintech Funnel Intelligence Platform
Business Problem Statement & Project Rationale

EXECUTIVE SUMMARY

RavenStack SaaS serves 500 B2B accounts across 5 industries with a \$11.3M MRR portfolio. A 22% account-level churn rate and fragmented pipeline visibility are causing significant, preventable revenue leakage. FinFlow Analytics was built to diagnose and resolve this.

KEY BUSINESS METRICS AT A GLANCE

Total Accounts	500	B2B companies
Account Churn Rate	22.0%	110 churned accounts
Total MRR	\$11.34M	Avg \$2,268 / subscription
Churn Events Logged	600	Top reason: Features (19%)
Avg Support Resolution	35.9 hrs	26% tickets urgent

1. Business Context & Industry Background

RavenStack is a B2B SaaS platform operating in the fintech-adjacent software space, serving companies across FinTech, DevTools, Cybersecurity, HealthTech, and EdTech verticals. The platform follows a subscription-based monetization model with three plan tiers (Basic, Pro, Enterprise) and generates revenue through monthly and annual billing cycles.

Industry Landscape

- SaaS companies globally average a churn rate of 5–7% annually; monthly churn above 2% is considered a crisis signal
- RavenStack's observed 22% account-level churn far exceeds industry benchmarks, representing a critical business risk
- Research shows acquiring a new customer costs 5–7x more than retaining an existing one — making churn reduction the highest-ROI initiative
- Funnel visibility gaps cost SaaS companies an estimated 15–25% of potential MRR annually through missed upsells and undetected at-risk accounts

RavenStack's Current Customer Portfolio

Industry	Accounts	Share	Churn Risk
DevTools	113	22.6%	Medium
FinTech	112	22.4%	High
Cybersecurity	100	20.0%	Low
HealthTech	96	19.2%	Medium
EdTech	79	15.8%	High

2. Core Business Problem

RavenStack lacks a unified, real-time intelligence layer that connects user acquisition, product engagement, subscription health, and churn signals into a single actionable view. As a result, the business is operating reactively — discovering churn after it has already occurred rather than intervening while there is still time.

2.1 The Five Pain Points

PAIN 1	22% of accounts have churned. With no real-time funnel tracking, the team cannot identify which stage in the customer journey accounts are dropping out of — trial, onboarding, or mature subscription. Without this visibility, retention campaigns are untargeted and ineffective.
PAIN 2	RavenStack has no mechanism to determine which features drive long-term retention vs. which are underused and causing frustration. With 25,000 usage events across 5,000 subscriptions, this data exists but has never been synthesized into insight.
PAIN 3	600 churn events cite reasons including features (19%), support quality (17.3%), budget (17.3%), and competitor alternatives (15.3%). Yet these signals are not being connected back to product or support decisions to create closed feedback loops.
PAIN 4	With 2,000 support tickets averaging 35.9 hours to resolve — and 26% marked urgent — support quality is a key churn driver. Yet no pipeline exists to surface which accounts experiencing poor support are also showing subscription risk signals.
PAIN 5	With \$11.34M in MRR across 5,000 subscriptions at varying plan tiers and billing frequencies, the business cannot accurately forecast revenue risk or quantify the financial impact of funnel drop-off without a structured analytics pipeline.

3. Business Impact Analysis

3.1 Revenue at Risk

Risk Category	Estimated Impact	Data Signal
Active Churn MRR Loss	~\$2.49M/yr	22% × \$11.34M MRR
Missed Upsell Revenue	~\$1.1M/yr	Upgrade flag = 12% of subs
Support Cost Overrun	High	35.9h avg resolution, 26% urgent
Refund Liability	\$8,652 logged	Avg \$14.42 per churn event

3.2 Churn Root Cause Breakdown

Churn Reason	Count	% of Total	Action Required
Feature Gaps	114	19.0%	Product roadmap realignment
Support Failure	104	17.3%	SLA enforcement + escalation
Budget Constraints	104	17.3%	Pricing tier optimization
Unknown Reason	95	15.8%	Exit survey implementation
Competitor Switch	92	15.3%	Competitive intelligence
Pricing Issues	91	15.2%	Value communication strategy

4. The FinFlow Analytics Solution

FinFlow Analytics is an end-to-end real-time funnel intelligence platform built to solve each of the five identified pain points through a structured data engineering and analytics pipeline.

4.1 Solution Architecture

Layer	Component	Purpose
Ingestion	Python Data Producer	Simulates real-time signup, subscription, usage, and churn events
Storage	MySQL (5 tables)	Persistent, relational storage of all event types across the funnel
Transform	SQL Pipeline	Aggregates raw events into realtime_funnel view with stage classification
Analytics	Python + Pandas	Computes KPIs: conversion rate, churn rate, usage depth, support correlation
Visualization	Matplotlib + Seaborn	5-panel dashboard: funnel, conversion, churn, usage by stage, correlation

4.2 How It Solves Each Pain Point

Pain Point	FinFlow Solution
No funnel visibility	Classifies every account into Trial / Active / Churned funnel stage in real time
Feature blindness	Aggregates usage_count & usage_duration_secs per stage to identify sticky vs. dead features
Disconnected churn signals	Links churn_reason_code to subscription and account data for root-cause attribution
Support-churn gap	Correlates resolution_time_hours and satisfaction_score with churn_flag via heatmap
Revenue forecasting gap	Tracks MRR, ARR, upgrade/downgrade flags to compute revenue risk by cohort

5. Funnel Stage Framework

FinFlow maps every account to one of four sequential funnel stages. Each stage has clear entry criteria, risk signals, and recommended interventions:

Stage	Status	Entry Criteria	Key Action
1. Trial	 Evaluation	is_trial = TRUE	Onboarding nudge, feature walkthrough
2. Active	 Retained	churn_flag = FALSE, trial = FALSE	Upsell to higher tier, engagement deepen
3. At-Risk	 Warning	Downgrade flag OR low usage_count	Proactive outreach, discount offer
4. Churned	 Lost	churn_flag = TRUE	Exit survey, win-back campaign

6. Data Assets & Schema

Dataset Summary

Dataset	Records	Key Columns	Business Role
ravenstack_accounts	500	churn_flag, plan_tier, industry	Master account registry & churn label
ravenstack_subscriptions	5,000	mrr_amount, upgrade_flag, churn_flag	Revenue & plan lifecycle tracking
ravenstack_feature_usage	25,000	usage_count, usage_duration_secs	Product engagement depth
ravenstack_churn_events	600	reason_code, refund_amount_usd	Churn root-cause attribution

Dataset	Records	Key Columns	Business Role
ravenstack_support_tickets	2,000	resolution_time_hours, priority	Support quality & escalation signals

7. Success Metrics & Expected Outcomes

7.1 Analytical Deliverables

- Funnel distribution chart showing account counts across Trial → Active → At-Risk → Churned
- Conversion rate metric: % of trial accounts converting to paid subscriptions
- Churn rate breakdown by reason code, plan tier, and industry vertical
- Feature usage heatmap: usage_count and usage_duration_secs segmented by funnel stage
- Support-churn correlation matrix linking ticket priority and resolution time to churn probability

7.2 Business KPIs to Monitor Post-Deployment

KPI	Current Baseline	Target (6 months)
Account Churn Rate	22.0%	< 15% (32% reduction)
Subscription Churn Rate	9.7%	< 6% (38% reduction)
Support Avg Resolution Time	35.9 hours	< 24 hours
Trial-to-Paid Conversion	Unmeasured	Establish baseline + 10% QoQ growth
MRR at Risk Identified	Unmeasured	> 80% of at-risk MRR flagged 30d before churn

8. Future Roadmap

Phase	Timeline	Enhancement
v2.0	Q3 2025	Apache Kafka integration for true real-time event streaming
v2.1	Q3 2025	Streamlit interactive dashboard replacing static matplotlib charts
v3.0	Q4 2025	ML churn prediction model (XGBoost/Random Forest) with 30-day early warning
v3.1	Q1 2026	NLP-based feedback classification on churn feedback_text field
v4.0	Q2 2026	Automated alert system: Slack/email triggers on at-risk account detection

9. Resume-Ready Project Description

Short Version (LinkedIn / Resume Bullet)
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Built FinFlow Analytics — a real-time SaaS funnel intelligence platform using Python, MySQL, Pandas, and Matplotlib — processing 25,000+ feature usage events across 500 B2B accounts to identify churn drivers, optimize conversion, and surface \$2.4M+ in at-risk MRR.

Extended Version (Portfolio / Project Description)

Designed and implemented an end-to-end real-time funnel analytics pipeline for a fintech SaaS platform with \$11.3M MRR. Built a MySQL event-stream architecture ingesting 5 data sources (500 accounts, 5,000 subscriptions, 25,000 usage events, 600 churn records, 2,000 support tickets). Developed SQL transformation pipelines to classify accounts into funnel stages and Python analytics modules to compute conversion rates, churn root-cause distributions, feature adoption scores, and support-churn correlations. Delivered a 5-panel executive visualization dashboard identifying 22% account churn, with actionable recommendations targeting a 32% churn reduction.